

Database Architecture

Centralized, Parallel, Distributed, and Application Architectures

1.7 Database Architecture

Definition

Database architecture refers to the design and organization of database systems, including hardware, software, and distribution of data.

Types of Database Architectures

- **Centralized Databases:**
 - Runs on one or a few cores with shared memory.
- **Client-Server:**
 - One server executes work on behalf of multiple clients.
- **Parallel Databases:** Exploit parallel computer architecture
 - Many-core shared memory
 - Shared disk
 - Shared-nothing architecture
- **Distributed Databases:**
 - Geographically distributed databases
 - May have schema and data heterogeneity

Centralized / Shared-Memory Architecture

Note

Most database users today access the system remotely over a network. **Client machines** run user applications, while **server machines** run the database system.

Two-Tier Architecture

- Application resides on the client machine.
- Client sends queries to the server using query languages (e.g., SQL).
- Communication often via APIs or standards such as ODBC and JDBC.

Three-Tier Architecture

- Client machine acts as a front-end only (no direct DB calls).
- Client communicates with an **application server** (often through a forms interface).
- Application server interacts with the database system to access data.
- Suitable for large web applications.

Example

Comparison: Two-tier: simpler, direct client-server communication. Three-tier: adds an intermediate application server for scalability, security, and modularity.